Study Questions: Agents — The Pattern-Reader's Mind

1. When you first pick up a new crochet pattern, what predictions does your mind generate before you make a single stitch? List at least three.

2. How does an experienced crocheter's generative model differ from a beginner's when reading the same pattern? Give a specific example involving a stitch abbreviation.

3. Explain why the instruction "work in pattern as established" is only understandable if the agent has built a sufficient internal model. What does this phrase require of the reader?

4. In what sense is reading a pattern an act of **active inference** rather than passive instruction-following?

5. Describe a time when your prediction about how a project would look or feel did not match the reality. What was the source of the prediction error — your model or the pattern?

6. How does looking at a finished-object photo before starting a project change your generative model compared to working from text-only instructions?

7. What are the **sensory states** available to you as a pattern-reading agent? List at least five sources of information you draw on while crocheting.

8. What are the **active states** you can deploy as a pattern-reading agent? List at least four actions you can take in response to prediction error.

9. A beginner asks you what "ch 3 (counts as dc)" means. What prior knowledge does your generative model contain that theirs does not?

10. Why might two experienced crocheters interpret the same ambiguous instruction differently? What does this reveal about the nature of generative models?

11. Describe the difference between "reading row by row" and "reading the pattern as a whole." How do these two approaches engage different levels of the generative model?
12. When you frog (rip out) a section of your work, what role does this play in the perception-action loop of active inference?
13. How does your self-model ("I am good at amigurumi but struggle with lace") affect which patterns you choose and how you approach them?
14. A pattern includes a schematic but no row-by-row photos. What must your generative model fill in to bridge the gap between the schematic and the stitch instructions?
15. What prediction errors would a crocheter experience if they skipped reading the "Notes" section of a pattern and went straight to Row 1?
16. How does the experience of "getting into a rhythm" with a pattern relate to the agent's generative model becoming well-calibrated?
17. Explain why a pattern that is highly rated by advanced crocheters might be poorly rated by beginners, even if it is objectively well-written. Frame your answer in terms of agent-model mismatch.
18. When you "swatch" for a new project, you are acting as an agent gathering sensory data. How does this information update your generative model before the project begins?
19. Consider a crochet-along where an instructor releases one section per week. How does the limited information change the agent's ability to generate predictions about the finished piece?
20. If you were designing a pattern specifically for beginner agents, what features would you include to compensate for their sparse generative models?